

AFSANEH CADEN, MSc

Data Scientist/Analyst

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I am a Data Scientist with a strong foundation in machine learning, deep learning, and NLP. I am eager to apply my skills to solve real-world problems. My background in telecommunications engineering has sharpened my analytical mindset, while my hands-on experience with large language models (LLMs), predictive analytics, and data visualization allows me to extract actionable insights from complex data.

KEY SKILLS

Programming & Databases: Python, R, SQL

Data Analysis & Visualization: pandas, NumPy, matplotlib, seaborn, Plotly, ggplot2, Tableau, Streamlit, ArcGIS

Machine Learning & NLP: scikit-learn, TensorFlow, PyTorch, NLTK, spaCy, Hugging Face (DistilBERT, BERT, RoBERTa), model fine-tuning

Cloud Computing: AWS (Pursuing AWS Cloud Practitioner Certification)

PROFESSIONAL EXPERIENCE

Data Scientist

01/2023 – 05/2024

Middle Tennessee State University (MTSU) – MSc

Murfreesboro, TN

- **Toxic Comment Classification** (LLMs, PyTorch, TensorFlow, Hugging Face)
 - Engineered and optimized a multi-headed deep learning model to classify six types of toxicity in Wikipedia's Talk Pages comments, achieving an accuracy of 98.8%.
 - Enhanced automated detection of harmful content improves platform moderation and fosters constructive interactions.
- **Business Intelligence Dashboard** (Tableau)
 - Developed an interactive BI dashboard to analyze product performance, market demand, and profitability trends, facilitating data-driven decisions through KPIs, geospatial insights, and customer activity analysis.

Data Scientist

01/2022 – 08/2022

Middle Tennessee State University (MTSU) – Graduate Certificate

Murfreesboro, TN

- **Heart Failure Mortality Prediction**
 - Developed a predictive model to identify critical survival factors in heart failure patients, complemented by a Streamlit dashboard app for real-time insights.

Private Math Tutor

05/2002 – 06/2016

Independent

Remote & in-person

- Instructed on various math topics, including algebra, trigonometry, and calculus.

Senior Telecom Engineer

07/2003 – 05/2011

Telecommunication Company of Iran

Tehran, Iran

- Optimized digital transmission networks by analyzing traffic patterns and system performance to enhance efficiency.
- Improved resource allocation and network reliability through predictive analytics and data-driven decision-making.
- Collaborated with cross-functional teams to optimize routing, signaling, and capacity planning, reducing downtime and improving service quality.

EDUCATION

Middle Tennessee State University

01/2023 – 05/2024

Data Science, Master of Science (MSc)

Murfreesboro, TN

Middle Tennessee State University

01/2022 – 08/2022

Data Science, Graduate Certificate

Murfreesboro, TN

Najaf Abad University

09/1997 – 06/2001

Telecommunication Engineering (BS)

Isfahan, Iran